SHEAFIFICATION TURNS HYPERCOVERING CONDITIONS INTO SURJECTIONS OF F(K)

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 684-708 of sources/stacks-hypercovering/pinned-original-source.tex; proof span is lines 710-760, the complete author proof (49 body lines).

Editorial note. Navigation stubs: 8 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 8 of queue_hypercovering.json (pb=2035, stmt 684-708, proof 710-760).

Editorial note. stacks-hypercovering wave-1 (hc01): fresh adjudication accept (2026-10-14); pool per standing fallback order (hypercovering -> pic -> spaces-more-cohomology) after md01 closed the moduli pool; hypercovering.tex 53 labels/34 with proof judged 26 accept / 8 reject / 14 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed. Named omission: Connectedness of (Delta/[m])_{<=n} for n >= 1 is left as an 'amusing exercise' - a named small fact; the F'/finite-limits argument is in-span.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 002S: sources/stacks-hypercovering/pinned-original-source.tex, source lines 760–760. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 002T: sources/stacks-hypercovering/pinned-original-source.tex, source lines 760–760. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01G3: sources/stacks-hypercovering/pinned-original-source.tex, source lines 510–521. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01G2: sources/stacks-hypercovering/pinned-original-source.tex, source lines 391–404. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:hypercovering.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0182: sources/stacks-hypercovering/pinned-original-source.tex, source lines 760–760. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0183: sources/stacks-hypercovering/pinned-original-source.tex, source lines 760–760. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 017Z: sources/stacks-hypercovering/pinned-original-source.tex, source lines 760–760. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00WT: sources/stacks-hypercovering/pinned-original-source.tex, source lines 760–760. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\mathcal{C}$ be a site with fibre products. Let $X \in \mathrm{Ob}(\mathcal{C})$ be an object of $\mathcal{C}$. Let K be a hypercovering of X. Consider the simplicial object $F(K)$ of $PSh(\mathcal{C})$, endowed with its augmentation to the constant simplicial presheaf h_X.*

1. *The morphism of presheaves $F(K)_0 \to h_X$ becomes a surjection after sheafification.*
2. *The morphism*
$$(d_0^1, d_1^1) : F(K)_1 \longrightarrow F(K)_0 \times_{h_X} F(K)_0$$
becomes a surjection after sheafification.
3. *For every $n \geq 1$ the morphism*
$$F(K)_{n+1} \longrightarrow (cosk_n sk_n F(K))_{n+1}$$
turns into a surjection after sheafification.

Proof. We will use the fact that if $\{U_i \to U\}_{i \in I}$ is a covering of the site $\mathcal{C}$, then the morphism
$$\amalg_{i \in I} h_{U_i} \to h_U$$
becomes surjective after sheafification, see Sites, Lemma 00WT. Thus the first assertion follows immediately.

For the second assertion, note that according to Simplicial, Example 0182 the simplicial object $\mathrm{cosk}_0\mathrm{sk}_0 K$ has terms $K_0 \times \ldots \times K_0$. Thus according to the definition of a hypercovering we see that $(d_0^1, d_1^1) : K_1 \to K_0 \times K_0$ is a covering. Hence (2) follows from the claim above and the fact that F transforms products into fibred products over h_X.

For the third, we claim that $\mathrm{cosk}_n\mathrm{sk}_n F(K) = F(\mathrm{cosk}_n\mathrm{sk}_n K)$ for $n \geq 1$. To prove this, denote temporarily F' the functor $\mathrm{SR}(\mathcal{C}, X) \to PSh(\mathcal{C})/h_X$. By Lemma 01G2 the functor F' commutes with finite limits. By our description of the cosk_n functor in Simplicial, Section 017Z we see that $\mathrm{cosk}_n\mathrm{sk}_n F'(K) = F'(\mathrm{cosk}_n\mathrm{sk}_n K)$. Recall that the category used in the description of $(\mathrm{cosk}_n U)_m$ in Simplicial, Lemma 0183 is the category $(\Delta/[m])^{opp}_{\leq n}$. It is an amusing exercise to show that $(\Delta/[m])_{\leq n}$ is a connected category (see Categories, Definition 002S) as soon as $n \geq 1$. Hence, Categories, Lemma 002T shows that $\mathrm{cosk}_n\mathrm{sk}_n F'(K) = \mathrm{cosk}_n\mathrm{sk}_n F(K)$. Whence the claim. Property (2) follows from this, because now we see that the morphism in (2) is the result of applying the functor F to a covering as in Definition 01G3, and the result follows from the first fact mentioned in this proof. □